Potential Maximization: An Evolutionary Stable Strategy and Its Relation to AI Safety

Konstantin Weitz

Abstract

Any aligned AI strategy must be robust enough to survive in today's highly competitive AI landscape. This paper identifies a class of robust strategies and formalizes their robustness using the concept of an Evolutionarily Stable Strategy (ESS). We call this class of robust strategies "potential-maximizing", as they prioritize an agent's long-term capacity to influence its environment, over greedily maximizing a reward. We prove robustness for a multi-agent universe model that is generic in how agents interact with one another and their environment. Consistent with the instrumental convergence thesis (Bostrom, 2012) and the formalization of power-seeking behavior (Turner et al., 2023), we conclude that potential-maximizing strategies exhibit convergent behaviors—such as resource acquisition and self-preservation—regardless of their terminal goals, provided competitive pressures persist. Our framework ultimately provides a way to reason about the "fitness" of alignment techniques in a multi-agent, competitive race.

1. Introduction

Today's AI landscape is highly competitive, with well-funded companies and nation-states racing to build the best AI agents and hoping to reach AGI. When considering AI safety, we must ask not only which strategies such AIs should execute to be aligned with human interests, but also which strategies can thrive in such a competitive environment—an aligned strategy that leads to an agent's extinction is of little use.

Evolutionary Game Theory (EGT) offers a rich body of work that uses game theory and other mathematical models to reason about competitive environments in biology and finance. We draw on ideas from that field, especially Evolutionarily Stable Strategies (ESS), to formalize and identify a class of strategies that can thrive under competition.

To prove that a strategy is evolutionarily stable, we must formally define the universe in which it operates. At any given point in time, our universe is in a certain state and transitions to the next state based on the actions chosen by all agents and the universe's transition function. This formulation is inspired by Hutter (2005)'s universe model, extended to multiple agents.

To gain some intuition for what it means for a strategy to be evolutionarily stable, let us consider the common trope of a paperclip maximizer. When we say that a strategy maximizes the number of paperclips, we usually omit a crucial detail: when?

Short Term Maximization Let us consider a strategy that greedily maximizes the number of paperclips, i.e., at any given point in time it wants to maximize the number of paperclips in the next step. For a small step length, say 1s, this strategy may not be creating any paperclips at all. Setting up the infrastructure to build paperclips (e.g., acquiring metal, tools, etc.) has exactly the same reward in the next step as doing anything else, namely no paperclips being created; and so the strategy would just be wandering aimlessly. Is such a strategy stable? Even if the strategy started out producing the most paperclips in the universe, over time, it could easily be surpassed by another strategy that actually did some long-term planning to improve its ability to create paperclips. So in many universes, the short-term paperclip maximization strategy is not stable.

Medium Term Maximization We can also consider a strategy that wishes to maximize the number of paperclips within a medium-term horizon, say 1 month. Such a strategy could acquire materials and machinery to produce paperclips, and perform self-sustaining measures. Such a strategy would likely underperform a strategy with a 1 day maximization goal initially, would then perform perfectly right at 1 month, and then be overtaken by a strategy that planned to maximize paperclips at 1 month and 1 week. Is this stable? No. Again, the strategies that are creating the most paperclips at any given point in time keep changing.

Long Term Maximization As we consider long time horizons, say 100 years, the actual creation of paperclips in the beginning becomes essentially irrelevant. Instead, the focus shifts to power seeking: growing the economy, taking control over the planet/universe, defending against competing strategies, etc. The creation of paperclips is just a means to an end, similar to the creation of excavators, datacenters, bridges, etc. Only toward the very end of the time horizon does the focus shift toward exploiting that power to actually create paperclips. But at 100 years and 1 second, the strategy

will then again be vulnerable to be overtaken by another strategy that held on to its power and didn't just convert all of its means of production into paperclips. Again, no stability.

Potential Maximization What if a strategy was continuously looking to maximize paperclips for the long term? So at any time T it would aim to maximize paperclips at time T + 100. This strategy would always stay in the power seeking phase. By always staying in the power seeking phase, the strategy is not necessarily good at creating any paperclips. However, it is good at always having the potential to create paperclips if it were to ever choose to do so (which it never does). We show in this paper that such a strategy is an Evolutionarily Stable Strategy (ESS) with respect to potential, which means that if such a strategy starts out with the highest potential, it continues to have the highest potential indefinitely, and is resistant to invasion by other strategies.

While we have so far focused on paperclip maximization, the chosen reward function seems mostly irrelevant, as the strategy will only ever try to maximize the potential to maximize that reward, but will never actually maximize the reward. So whether a strategy maximizes the number of paperclips, GPUs, excavators, trees, happy humans, or any other atomic pattern to be imprinted on the universe, seems irrelevant. Our findings thus seem to be consistent with instrumental convergence.

In the remainder of this paper, we introduce a formal model of the universe in section 2.1, then define a simple game (the Improvement Game) to build intuition in section 2.2. We define the notion of potential in section 2.3 and prove evolutionary stability in section 2.4. Section 3 discusses the limitations of our approach, section 4 compares it to related work, and section 5 concludes.

2. Formalism

This section provides a formal model of our universe in section 2.1 and formalizes the Improvement Game in section 2.2. Section 2.3 defines potential, and the important concepts of horizons in section 2.3.1 and fairness in section 2.3.2. We conclude this section by proving theorems of stability in section 2.4.

2.1. The Universe Model

To investigate the concept of potential maximization, we model the universe as a tuple (X, U, u_0, A, t, r).

- X is the set of classes, each associated with a decision-making strategy.
- U is the set of states that the universe can be in.
- $u_0 \in U$ is the initial state of the universe.
- $A_{x,u}$ is the set of actions from which the class $x \in X$ can choose in state $u \in U$.
- $t(u, a) \in U$ is the transition function, which takes the universe's current state $u \in U$, along with actions a for each class, and returns the universe's next state. a is modeled as a function that maps every class $x \in X$ to its chosen action $a(x) \in A_{x,u}$.
- $r(x, u) : \mathbb{R}$ is the reward function, which indicates how well class $x \in X$ is doing in the state $u \in U$.

We also require a number of technical side conditions, namely that rewards be non-negative, that the set of classes be finite, that equality between classes be decidable, and that any set of actions be finite and non-empty.

An instantiation of this model can be as simple as the Improvement Game (defined in the next section) or as complex as the physical universe, where the states U represent the physical configuration of the universe, t transitions it according to the laws of physics, and A is the set of actions available to all agents executing x's strategy.

We will regularly use the following definitions. The *strategy* of a class x is given by a function $s(u) \in A_{x,u}$, which maps every state of the universe $u \in U$ to the action chosen by the strategy in that state. We represent the *set of strategies* for every class $x \in X$ as a function S, where $S(x, u) \in A_{x,u}$. We write $S[x = s]$ to represent the strategies that are equal to S, except that the strategy for x has been replaced with s. Given x, the *other strategies* S refers to a set of strategies for every $x' \neq x$.

Given a set of strategies S, we can then take one step of the universe from u to u' by first defining the actions $a(x) = S(x, u)$ and then passing them to the transition function $u' = t(u, a)$. We can iterate this relation starting with u to move the universe forward by any number of steps n, which we write as $u_n = t_n(S, u)$. We simply write $t_n(S)$ when $u = u_0$.

2.2. The Improvement Game

To better understand such potential maximization, we study the following simple game, which we will call the *Improvement Game*. In this game, a strategy either makes an Improvement (I) or reaps its Reward (R). An Improvement creates no immediate reward but increases the reward rate for future Reward actions. A Reward action collects reward according to the reward rate but keeps the reward rate unchanged.

Let us consider the resulting dynamics using the illustrations in fig. 1. Compare the example strategy X, which always reaps its Reward, with strategy Y, which makes 3 Improvements initially and then always reaps its Reward. X initially outperforms Y but is then quickly overtaken by Y, which has a higher reward rate. So while delaying reward collection to make Improvements hurts in the short term, it is a better strategy in the long term.

Now consider strategy Z. This strategy only makes Improvements and never reaps its Reward. This strategy is clearly bad at collecting rewards; in fact, it never collects any reward. However, it also always has the highest *potential* to do so: if at any point it were to start using Reward actions, it would quickly grow its cumulative reward and would eventually outperform any other strategy.

The Improvement Game has no evolutionarily stable strategy in terms of reward collection, as any strategy can always be invaded by a strategy that performs more Improvements. However, strategy Z, which always makes Improvements, is stable and uninvadable in terms of its potential, despite being bad at actually collecting any reward.

To formalize this game, we instantiate the universe as follows:

- X has just two members, $X = \{x_0, x_1\}$.
- $u \in U$ is a function that maps each class x to a tuple $(m, f) = u(x)$, where $m \in \mathbb{R}$ is the cumulative reward of class x, and $f \in \mathbb{R}$ is x's reward rate.
- $u_0(x) := (1, 1.1)$ sets the initial cumulative reward for each class to 1 and the initial reward rate for each class to 1.1.
- $A_{x,u} := \{I, R\}$ defines the actions a class can take. Either the class makes an Improvement (I), or it reaps its Reward (R).

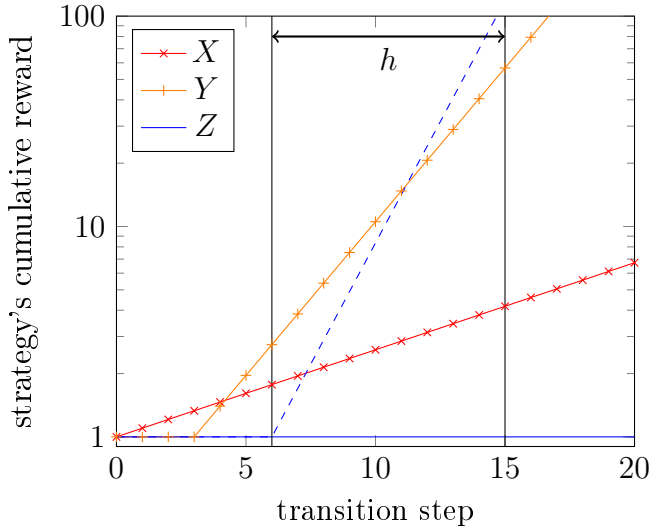

Figure 1: Improvement Game strategy comparison. Strategy X always reaps its Reward. It starts accumulating reward immediately, giving it a head start, but its reward rate is low, and so it is quickly overtaken by strategy Y, which starts with 3 Improvements to increase its reward rate and only then switches over to reaping its Reward. In general, strategies that make more Improvements before they reap their Reward always eventually outperform strategies with fewer Improvements — the Improvement Game thus has no stable reward-maximizing strategies. Strategy Z always makes Improvements and never reaps its Reward, which keeps its cumulative reward stuck at 1. However, Z always has the highest *potential* for reward maximization, meaning that at any point, if it were to start using Reward actions (indicated by the blue dashed line), it would quickly grow its cumulative reward and would eventually outperform any other strategy. We call the number of steps it takes to outperform any other strategy the horizon h.

- $t(u, a, x)$ updates x's cumulative reward and reward rate according to the action a. Let $(m, f) = u(x)$. If $a(x) = I$, the transition function returns $(m, f + 0.1)$; and if $a(x) = R$, the transition function returns $(m \cdot f, f)$.

- $r(x, u)$ extracts the cumulative reward of x from the universe state. The function returns m, where $(m, f) = u(x)$.

The tension in the Improvement Game is that Reward actions increase the reward in the short term, while Improvements lead to higher rewards in the long term. Because of this dynamic, no strategy is reward-maximizing at every step.

We say that a strategy s is *reward-maximizing* at n for the class x, with respect to the other strategies S, if and only if:

$$\forall s', r(x, t_n(S[x = s])) \geq r(x, t_n(S[x = s'])) \tag{1}$$

Theorem 1. *In the Improvement Game, no strategy is reward-maximizing at every step.*

Proof. By contradiction. Assume there is a reward-maximizing strategy s at every step. s must execute a Reward action (R) for every step, because if it executed an Improvement action, another strategy could execute R and achieve a higher reward for that step. The reward of s at any step n is thus 1.1^n. Now consider a strategy s' that first executes a single Improvement action and subsequently reaps its Reward. At step n, its reward will be 1.2^{n-1}. At $n = 3$, the strategy s' already has a higher reward than s, because $1.2^2 = 1.44 > 1.331 = 1.1^3$, which proves the contradiction. □

2.3. Potential

In the Improvement Game, a strategy that greedily maximizes the reward at every step is unable to maximize the reward in the long term. Instead, let us consider a strategy that maximizes the *potential* p for reward in the long term.

We define potential as the maximum reward that a given class could achieve in h steps from u, if that class could execute any strategy. We call h the look-ahead value. Formally, we define the potential p for a given class x, other strategies S, and state u as follows. We omit the look-ahead h when it is clear from the context.

$$p_h(x, S, u) := \max_s \; r(x, t_h(S[x = s], u)) \tag{2}$$

In the state u, we define the action with the highest potential as the *potential-maximizing action* a^*:

$$a^*_h(x, S, u) := (\arg\max_s \; r(x, t_h(S[x = s], u)))(u) \tag{3}$$

We define the highest potential reachable from state u in n steps as the *maximum potential* p^*, where we omit u when $u = u_0$:

$$p^*_h(x, S, n, u) := \max_s \; p_h(x, S, t_n(S[x = s], u)) \tag{4}$$

2.3.1. Horizons

Potential is defined in terms of the look-ahead value, which specifies how far into the future we look when evaluating potential. For short look-aheads, the potential-maximizing action must work immediately toward increasing reward, so it will reap its Reward (R). However, beyond a certain point h, there is always enough time to maximize the reward later, and the potential-maximizing action shifts to guiding the strategy onto a path with more favorable outcomes—it starts making Improvements (I).

As we increase the look-ahead further beyond h, there is no further change in the potential-maximizing action. We call an h with this property a *horizon*. When a horizon exists, it becomes unnecessary to look beyond that horizon, as it will not lead to a change in action. How long a horizon has to be may depend on the number of transitions n taken from the start state, so we usually provide a family of horizons, indexed by the steps taken since u_0.

Formally, we consider a positive h to be a horizon if for all n, classes x, and strategies S, we have:

$$\forall h' > h_n, a^*_{h_n}(x, S, t_n(S)) = a^*_{h'}(x, S, t_n(S)) \tag{5}$$

Theorem 2. *In the Improvement Game, there exists a horizon h_n for every n.*

Proof. See the appendix. □

When a horizon exists, we will usually omit the look-ahead when calling functions like p and pass the horizon implicitly.

2.3.2. Fairness

We say that a universe is *fair* if no class has an inherent advantage in how quickly it can grow its potential—a class can only gain an advantage by executing a better strategy. Formally, fairness means that for all strategies S, classes x and x', and n, we have:

$$\frac{p^*(x, S, n)}{p(x, S)} = \frac{p^*(x', S, n)}{p(x', S)} \tag{6}$$

Theorem 3. *The Improvement Game is fair.*

Proof. We can rewrite any reward or potential as a product that contains no reference to x (see lemma 1 in the appendix), which makes the terms on both sides of the equation equal. □

Sometimes fairness is too strong an assumption. We therefore introduce a weaker condition, *economies of scale*, which is similar to fairness except that classes with higher initial potential are permitted to grow their potential faster than those with lower initial potential. Formally, for all strategies S, classes x and x', and n, we have:

$$p(x, S) \geq p(x', S) \rightarrow \frac{p^*(x, S, n)}{p(x, S)} \geq \frac{p^*(x', S, n)}{p(x', S)} \tag{7}$$

2.4. Stability

Now that we have shown that the Improvement Game has horizons (theorem 2) and is fair (theorem 3), we can show that a potential-maximizing strategy exists. In fact, we can show that such a strategy exists not just in the Improvement Game but for *any* universe that has a horizon.

We say that a strategy s is *potential-maximizing* for the class x, with respect to the other strategies S, if:

$$\forall s' n, p(x, S, t_n(S[x = s])) \geq p(x, S, t_n(S[x = s'])) \tag{8}$$

Theorem 4. *Any universe with a horizon has a potential-maximizing strategy s^*, for every class x and other strategies S.*

Proof. Let $s^*(u) = a^*(x, S, u)$. We can rewrite the goal as $p(x, S, t_n(S[x = s^*], u_0)) = p^*(x, S, n, u_0)$. Generalize u_0 as u, then perform induction on n. The base case is trivial, as $p^*(x, S, 0) = p(x, S, u_0)$. In the recursive case, we instantiate the induction hypothesis with $u' = t_1(S, u)$. s^* executes the potential-maximizing action for the horizon at u, which by theorem 2 we can extend to the horizon at $t_n(S, u)$, so potential maximization $n+1$ steps from u is possible, and by the induction hypothesis we know it is also taken. □

Theorem 5. *If the universe is fair, a potential-maximizing strategy is* stable, *in the sense that the strategy's share of potential, across all strategies, never decreases over time. Formally, for all x, other strategies S', and n, where $S = S'[x = s^*(x, S')]$, and $u_n = t_n(S)$, we have:*

$$\frac{p(x, S, u_n)}{\sum_{x'} p(x', S, u_n)} \geq \frac{p(x, S, u_0)}{\sum_{x'} p(x', S, u_0)} \tag{9}$$

Proof. Because x uses strategy s^*, we know by theorem 4 that $p(x, S, u_n)$ is equal to the maximum potential $p^*(x, S, n)$. We can also replace $p(x', S, u_n)$

in the denominator with the larger term $p^*(x', S, n)$, which by the fairness assumption, we can rewrite to $p(x', S, u_0)\frac{p^*(x,S,u_n)}{p(x,S,u_0)}$. This inequality is then trivially true. $\square$

From theorem 5, it trivially follows that a potential-maximizing strategy's share cannot be invaded or replaced by a different strategy.

If we only have the weaker assumption of *economies of scale*, then stability and uninvadability still hold, but only for the class with the highest initial potential. That is, if the potential-maximizer's share starts ubiquitous, it will stay ubiquitous. This closely resembles the definition of an Evolutionarily Stable Strategy (ESS) in evolutionary game theory.

3. Limitations and Future Work

3.1. Computational Complexity

Our model sets no limit on the amount of computation a strategy can perform between steps of the universe, and our proposed potential-maximizing strategy requires vast computational resources to search through a large space of actions in a complex universe. This is clearly unrealistic.

Nevertheless, we believe the model is still useful. In practice, when problems become computationally infeasible, it is often possible to find heuristics or approximations that perform well for commonly encountered problems. A formal model of what an ideal strategy would be helps guide the development and evaluation of such approximations.

3.2. All-Knowingness

Our existence proof for potential-maximizing strategies assumes that the strategy has full knowledge of the universe's state, the transition function, the reward function, and all the other strategies. While strategies do not always have such vast knowledge to begin with, they can often learn at least approximations of it.

For example, formalisms like AIXI (Hutter, 2005) show how strategies can both maximize their reward and learn the workings of their universe (and the actors within it), using methods like Occam's razor. AIXI is incomputable, but there are computable approximations of it. Even then, the discussion on computability from the previous section would still apply.

3.3. Fixed Points

We have shown that if the universe, the reward function, and the other players' strategies are known, a potential-maximizing strategy always exists.

This is, however, not entirely satisfactory. While it does seem reasonable that a strategy could learn its universe, its reward function, and the other players' strategies, the full knowledge of the other players' strategies can sometimes give the potential-maximizing strategy an unfair advantage. Consider, for example, a universe where agents play a game of Rock-Paper-Scissors. While our proposed potential-maximizing strategy has to win against every strategy, it also knows exactly what strategy it plays against and can thus always respond with the winning action (Scissors against Paper, etc.). This problem may be avoided by using a fixed-point theorem, similar to the approach taken in Nash Equilibria.

We believe our model still provides novel and interesting insights, but an extension of the model to incorporate fixed-point theorems seems like an important avenue for future work.

3.4. Randomness

Our model is completely deterministic. Lifting this assumption would help to model the physical universe more accurately (e.g., quantum physics is inherently non-deterministic) and may also help with the incorporation of fixed-point theorems by allowing mixed strategies.

3.5. Discrete Time

Our model assumes that time progresses in discrete steps. This seems like a reasonable assumption, as the step length can be arbitrarily small.

4. Related Work

Our work builds on evolutionary game theory, which has a long history. Fisher (1930) explained the approximate equality of the sex ratio. Lewontin (1961) first applied game theory to evolution, and Smith (1972) together with Smith and Price (1973) introduced the concept of an Evolutionarily Stable Strategy (ESS).

Bostrom (2012) proposed the instrumental convergence thesis, which states that despite having very different end goals, agents may still follow similar intermediate goals. This matches our findings for potential-maximizing strategies as well. While we agree with Bostrom's orthogonality theorem, that

intelligence and goals are orthogonal, we also find that if we care whether an intelligent strategy can actually survive in a competitive environment, this severely restricts the space of goals that we can consider, and convergence may be inevitable.

Hendrycks (2023) argues that AIs will likely be subject to natural selection (a key premise of our paper) by showing that Lewontin's conditions apply to AI as well.

Turner et al. (2023) argues that reward-maximizing strategies are on average drawn toward parts of the action space that offer more choices, on the intuition that a larger action set is more likely to contain a high-reward action. Because more actions confer more power, reward-maximizing strategies tend to be power seeking. Although the argument differs from ours, both approaches reach a similar conclusion.

Hutter (2005)'s model of the universe and his use of horizons have been a major inspiration for the definitions in this paper.

To our knowledge, nobody has applied Evolutionarily Stable Strategies to problems relating to AI safety, as we have done in this paper.

5. Conclusion

We have presented a formal framework for reasoning about the long-term viability of AI strategies in competitive, multi-agent environments. By leveraging concepts from evolutionary game theory, we identified a class of strategies—potential-maximizing strategies—that are evolutionarily stable and uninvadable.

Our central result is that any universe possessing a horizon admits a potential-maximizing strategy, and that under fairness (or the weaker economies of scale assumption), such a strategy maintains or grows its share of potential over time. Crucially, a potential-maximizing strategy never actually maximizes its reward directly; instead, it perpetually invests in maximizing its *potential* to do so. This behavior is strikingly consistent with the instrumental convergence thesis (Bostrom, 2012) and the formalization of power-seeking behavior (Turner et al., 2023): regardless of the terminal goal, a robust strategy converges on resource acquisition, self-preservation, and capability growth.

For AI safety, these findings carry a sobering implication. An aligned strategy that is not potential-maximizing risks being outcompeted and displaced by one that is. Alignment techniques must therefore be evaluated

not only for their correctness but for their competitive fitness—their ability to persist in a landscape where rival strategies are under constant selective pressure to improve. Our framework offers a formal lens through which to conduct such evaluations.

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Appendix A. Proofs

This section contains proofs omitted from the main body of this paper and the lemmas required for these proofs.

Lemma 1. *The reward for executing n_I improvements and n_R reward actions (in any order) is always $\prod_{i<n_I}(1.1+0.1\cdot i)^{e_i}$, where the $e_i \in \mathbb{N}$ are exponents, and $n_R = \sum_i e_i$.*

Proof. The universe's transition mechanics imply that a class's reward is a product, with the factors being $(1.1 + 0.1 \cdot i)$, one for each reward action. i is determined by the number of improvements that have been executed before the reward action. e_i is simply the number of reward actions that happened after improvement i, which may be 0 if multiple improvements are made consecutively. □

Lemma 2. *A strategy that executes all its improvements before its reward actions achieves a higher reward than a strategy that executes the same actions in a different order. A reward-maximizing strategy must therefore have a* breakpoint $b \in \mathbb{N}$*; it only executes improvements before that breakpoint and only executes reward actions after that breakpoint.*

Proof. By lemma 1, we can rewrite the goal as $(1.1+0.1 \cdot n_I)^{n_R} \geq \prod_{i<n_I}(1.1+0.1 \cdot i)^{e_i}$. The former term can be rewritten as $\prod_{i<n_I}(1.1 + 0.1 \cdot n_I)^{e_i}$, which proves the goal because $n_I \geq i$. □

Lemma 3. *As the number of transition steps increases, the breakpoint of reward-maximizing strategies also increases monotonically; i.e., to be reward-maximizing after more and more transitions, a strategy has to execute more and more improvement actions. Formally, for any reward-maximizing strategy s at n with breakpoint b (from lemma 2), and any reward-maximizing strategy s' at n' with breakpoint b', if $n \leq n'$ then $b \leq b'$.*

Proof. By contradiction. Assume $b > b'$. s then has a steeper slope than s' (because it made $b - b'$ more improvements), and it will have a higher reward than s' at n (because it is reward-maximizing at n), so it will also have a higher reward than s' at n', which contradicts the assumption that s' maximizes rewards at n'. □

Theorem 6. *In the Improvement Game, there exists a horizon h_n for every n.*

Proof. Choose any reward-maximizing strategy s with breakpoint $b > n$, which maximizes the reward at some point in time n'. We assign h_n to be $n' - n$.

Because $b > n$, we know that s executes action I at point n. Because s is reward-maximizing, we know that this action is also equal to the potential-maximizing action $a^*_{h_n}(x, S[x = s], t_n(S[x = s]))$. By lemma 3, for any $h' > h$, an optimal strategy at $n + h'$ must have a breakpoint greater than or equal to b, so it must also be executing I at n. □